

a)

The screenshot shows the AWS Management Console interface for creating a new key pair. The breadcrumb navigation at the top indicates the path: EC2 > Key pairs > Create key pair. The page title is 'Create key pair' with an 'Info' link. A descriptive paragraph explains that a key pair consists of a private key and a public key used for authentication. The form includes a 'Name' field with the value 'abra_Lab_1_Key' and a note about character limits. The 'Key pair type' section has 'RSA' selected. The 'Private key file format' section has '.pem' selected. There is an 'Add new tag' button and a 'Create key pair' button at the bottom right.

Create key pair [Info](#)

Key pair
A key pair, consisting of a private key and a public key, is a set of security credentials that you use to prove your identity when connecting to an instance.

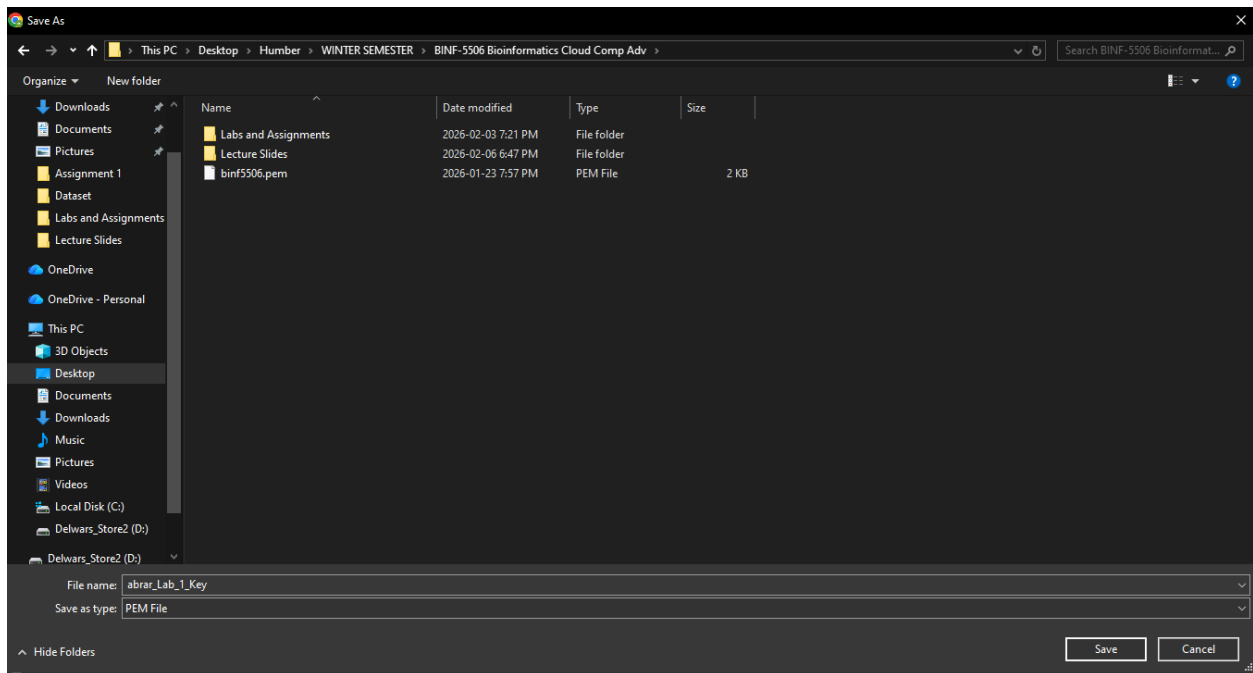
Name
abra_Lab_1_Key
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type [Info](#)
☒ RSA ☐ ED25519

Private key file format
☒ .pem
For use with OpenSSH
☐ .ppk
For use with PuTTY

Tags - optional
No tags associated with the resource.
[Add new tag](#)
You can add up to 50 more tags.

[Cancel](#) [Create key pair](#)



b)

EC2 > Instances > Launch an instance

Enable

Firewall (security groups)

Create security group

Select existing security group

We'll create a new security group called 'launch-wizard-2' with the following rules:

Allow SSH traffic from

Helps you connect to your instance

Anywhere

0.0.0.0/0

Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Configure storage

1x 3d GIB gp3 Root volume, 3000 IOPS, Not encrypted

Add new volume

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance.

Summary

Number of instances 1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-0b6c6ebed2801a5cb

Virtual server type (instance type)
t2.large

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 30 GiB

Cancel

Launch instance

Preview code

CloudShell Feedback

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aws Search [Alt+S]

United States (N. Virginia) Account ID: 0313-0511-0421 voclabs/user4544613-n10005099@humbar.ca

EC2 > Instances

Instances (1/2)

Find Instance by attribute or tag (case-sensitive)

All states

Last updated 8 minutes ago

Connect Instance state Actions Launch instances

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
☒	Abrar_Lab_1_...	i-097041921de0da7df	Running	t2.large	2/2 checks pass	View alarms +	us-east-1e	ec2-54-160-75-225.co...	54.160
☐	BINFS506	i-0cccd236bc3d55cf52	Running	t2.large	2/2 checks pass	View alarms +	us-east-1d	ec2-100-51-159-167.co...	100.51

i-097041921de0da7df (Abrar_Lab_1_Server)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary

Instance ID
i-097041921de0da7df

IPv6 address
-

Hostname type
IP name: ip-172-31-54-100.ec2.internal

Answer private resource DNS name

Public IPv4 address
54.160.75.225 | open address

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-54-100.ec2.internal

Instance type

Private IPv4 addresses
172.31.54.100

Public DNS
ec2-54-160-75-225.compute-1.amazonaws.com | open address

Elastic IP addresses

CloudShell Feedback

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i-097041921de0da7df (Abrar_Lab_1_Server)

Instance details

AMI ID
ami-0b6c6ebed2801a5cb

AMI name
ubuntu/images/hvm-ssd-gp3/ubuntu-noble-24.04-amd64-server-20251212

Stop protection
Disabled

Instance reboot migration
Default (On)

Monitoring
disabled

Allowed image
-

Launch time
Fri Feb 06 2026 19:48:29 GMT-0500 (Eastern Standard Time) (24 minutes)

Instance auto-recovery
Default

Platform details
Linux/UNIX

Termination protection
Disabled

AMI location
amazon/ubuntu/images/hvm-ssd-gp3/ubuntu-noble-24.04-amd64-server-20251212

Lifecycle
normal

c)

Successfully Initiated rebooting of i-097041921de0da7df

Instances (1/2) Info

Find Instance by attribute or tag (case-sensitive)

All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
Abrar_Lab_1...	i-097041921de0da7df	Running	t2.large	2/2 checks passed	View alarms +	us-east-1e	ec2-100-49-166-191.co...	100.49

i-097041921de0da7df (Abrar_Lab_1_Server)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID: i-097041921de0da7df

IPV6 address: -

Hostname type: IP name: ip-172-31-54-100.ec2.internal

Answer private resource DNS name

Public IPv4 address: 100.49.166.191 | open address

Instance state: Running

Private IPv4 addresses: 172.31.54.100

Public DNS: ec2-100-49-166-191.compute-1.amazonaws.com | open address

Private IP DNS name (IPv4 only): ip-172-31-54-100.ec2.internal

Elastic IP addresses

d)

File Edit Selection View Go Run ...

ubuntu [SSH: Abrar_Lab_1_Server]

EXPLORER

- UBUNTU [SSH: ABRAR_LAB_1_SERVER]
 - .cache
 - .ssh
 - .vscode-server
 - Bioinformatics_Cloud_Computing / lab_1
 - genomic_analysis_script.sh
 - .bash_history
 - .bash_logout
 - .bashrc
 - .profile
 - .wget-hsts

genomic_analysis_script.sh

```
1 #!/bin/bash
2 set -e
3
4 SRA=SRR1972739
5 REF_ID=AF086833.2
6 RESULTS_FOLDER=results
7 RAW_DIR=${RESULTS_FOLDER}/raw
8 ALIGNED_DIR=${RESULTS_FOLDER}/aligned
9 VARIANT_DIR=${RESULTS_FOLDER}/variants
10 ANNOTATED_DIR=${RESULTS_FOLDER}/annotated
11 QC_DIR=${RESULTS_FOLDER}/qc
12 SNPEFF_DIR=${RESULTS_FOLDER}/snpeff
13 SNPEFF_DATA_DIR=${SNPEFF_DIR}/data/reference_db
14
15 mkdir -p $RAW_DIR $ALIGNED_DIR $VARIANT_DIR $ANNOTATED_DIR $QC_DIR $SNPEFF_DATA_DIR
16
17 echo Downloading reference genome...
18 efetech -db nucleotide -id $REF_ID -format fasta > $RAW_DIR/reference.fasta
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

To run a command as administrator (user "root"), use "sudo <command>". See "man sudo_root" for details.

```
ubuntu@ip-172-31-54-100:~$ git clone https://github.com/SohailHanif/Bioinformatics_Cloud_Computing.git
Cloning into 'Bioinformatics_Cloud_Computing'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (2/2), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 2 (from 1)
Receiving objects: 100% (4/4), done.
ubuntu@ip-172-31-54-100:~$
```

SSH: Abrar_Lab_1_Server master 0 0 0 0 Ln 1, Col 1 Spaces: 2 UTF-8 LF {} Shell Script Finish Setup

e)

```
PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS
bash - ubuntu + - [ ] [ ] ... | [ ] [ ] X

ubuntu@ip-172-31-54-100:~$ sudo apt install tree
Preparing to unpack .../tree_2.1.1-2ubuntu3.24.04.2_amd64.deb ...
Unpacking tree (2.1.1-2ubuntu3.24.04.2) ...
Setting up tree (2.1.1-2ubuntu3.24.04.2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@ttyS0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
ubuntu @ session #7: node[1258,1297], sh[1254]
ubuntu @ session #9: code-bdd88df003[1432], sshd[1363,1413]
ubuntu @ user manager service: systemd[1094]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-54-100:~$ tree
.
├── Bioinformatics_Cloud_Computing
│   └── lab_1
│       └── genomic_analysis_script.sh
└──

3 directories, 1 file
```

f)

```
14 wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
15 bash Miniconda3-latest-Linux-x86_64.sh
16 conda create --prefix ./bioenv
17 conda activate ./bioenv
18 conda config --set channel_priority strict
19 conda config --add channels bioconda
20 conda config --prepend channels conda-forge
21 conda install entrez-direct sra-tools fastqc samtools bwa gatk4 snpEff -y
```

```
(/home/ubuntu/bioenv) ubuntu@ip-172-31-54-100:~$ conda list
```

```
# packages in environment at /home/ubuntu/bioenv:
```

```
#
```

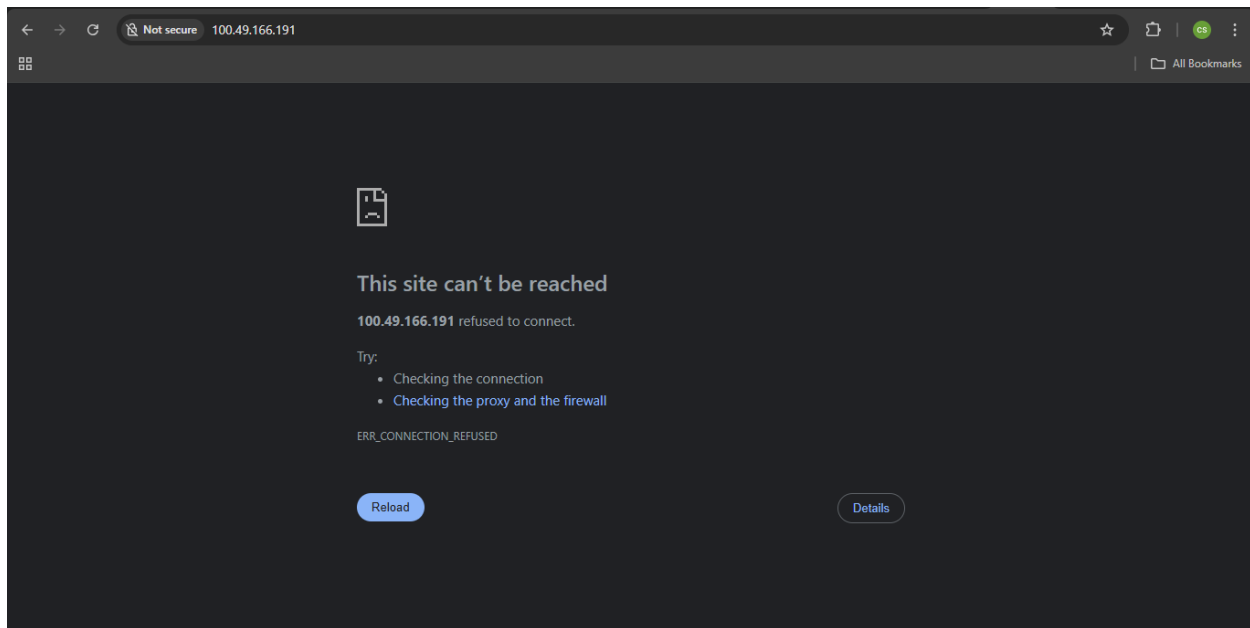
# Name	Version	Build	Channel
_libgcc_mutex	0.1	conda_forge	conda-forge
_openmp_mutex	4.5	2_gnu	conda-forge
alsa-lib	1.2.15.3	hb03c661_0	conda-forge
bwa	0.7.19	h577a1d6_1	bioconda
bzip2	1.0.8	hda65f42_8	conda-forge
c-ares	1.34.6	hb03c661_0	conda-forge
ca-certificates	2026.1.4	hbd8a1cb_0	conda-forge
cairo	1.18.4	he90730b_1	conda-forge
curl	8.18.0	h4e3cde8_0	conda-forge
entrez-direct	24.0	he881be0_0	bioconda
fastqc	0.12.1	hdfd78af_0	bioconda
font-ttf-dejavu-sans-mono	2.37	hab24e00_0	conda-forge
font-ttf-inconsolata	3.000	h77eed37_0	conda-forge
font-ttf-source-code-pro	2.038	h77eed37_0	conda-forge
font-ttf-ubuntu	0.83	h77eed37_3	conda-forge
fontconfig	2.15.0	h7e30c49_1	conda-forge
fonts-conda-ecosystem	1	0	conda-forge
fonts-conda-forge	1	hc364b38_1	conda-forge
freetype	2.14.1	ha770c72_0	conda-forge
gatk4	4.6.2.0	py310hdfd78af_1	bioconda
giflib	5.2.2	hd590300_0	conda-forge
graphite2	1.3.14	hecca717_2	conda-forge
harfbuzz	12.3.2	h6083320_0	conda-forge
htslib	1.23	h566b1c6_0	bioconda
icu	78.2	h33c6efd_0	conda-forge
keyutils	1.6.3	hb9d3cd8_0	conda-forge
krb5	1.21.3	h659f571_0	conda-forge
lcms2	2.17	h717163a_0	conda-forge
ld_impl_linux-64	2.45.1	default_hbd61a6d_101	conda-forge
lerc	4.0.0	h0aef613_1	conda-forge
libcups	2.3.3	hb8b1518_5	conda-forge
libcurl	8.18.0	h4e3cde8_0	conda-forge
libdeflate	1.22	hb9d3cd8_0	conda-forge

```
(/home/ubuntu/bioenv) ubuntu@ip-172-31-54-100:~/Bioinformatics_Cloud_Computing/lab_1$ bash genomic_analysis_script.sh
```

```
Pipeline completed successfully! Check the folders in results for output files.
```

```
.
├── genomic_analysis_script.sh
├── results
│   ├── aligned
│   │   ├── aligned.sam
│   │   ├── aligned.sorted.bam
│   │   ├── dedup.bam
│   │   ├── dedup.bam.bai
│   │   └── dup_metrics.txt
│   ├── annotated
│   │   └── annotated_variants.vcf
│   ├── qc
│   │   ├── SRR1972739_fastqc.html
│   │   └── SRR1972739_fastqc.zip
│   ├── raw
│   │   ├── SRR1972739
│   │   │   └── SRR1972739.sra
│   │   ├── SRR1972739.fastq
│   │   ├── reference.dict
│   │   ├── reference.fasta
│   │   ├── reference.fasta.amb
│   │   ├── reference.fasta.ann
│   │   ├── reference.fasta.bwt
│   │   ├── reference.fasta.fai
│   │   ├── reference.fasta.pac
│   │   └── reference.fasta.sa
│   ├── snpEff
│   │   ├── data
│   │   │   ├── reference_db
│   │   │   │   ├── genes.gbk
│   │   │   │   └── sequence.bin
│   │   │   └── snpEffectPredictor.bin
│   │   ├── snpEff.config
│   │   ├── snpEff.genes.txt
│   │   └── snpEff.html
```

g)



Index of /

Name	Last modified	Size	Description
------	---------------	------	-------------

 lab_1/	2026-02-07 02:16	-	
--	------------------	---	--

Index of /lab_1/results

Name	Last modified	Size	Description
------	---------------	------	-------------

 Parent Directory		-	
 aligned/	2026-02-07 02:16	-	
 annotated/	2026-02-07 02:16	-	
 qc/	2026-02-07 02:16	-	
 raw/	2026-02-07 02:16	-	
 snpEff/	2026-02-07 02:16	-	
 variants/	2026-02-07 02:16	-	

h) *I accidentally released the elastic IP instead of Disassociate

The image consists of two screenshots of the AWS Management Console, illustrating a mistake in managing Elastic IP addresses.

Top Screenshot: Elastic IP addresses

- Header:** Elastic IP address associated successfully. Elastic IP address 54.86.29.113 has been associated with Instance I-0c0d236bc3d55cf52.
- Table:** Elastic IP addresses (1/2)

Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record	Associated instance ID
Abrar_elip_lab_1	100.51.159.167	Public IP	eipalloc-08cb2503c496b47e	-	-
-	54.86.29.113	Public IP	eipalloc-03c5612cd799fadc1	-	I-0c0d236bc3d55cf52

- Summary:** View IP address usage and recommendations to release unused IPs with [Public IP insights](#).

Bottom Screenshot: EC2 Instances

- Header:** Successfully initiated termination (deletion) of i-097041921de0da7df.
- Table:** Instances (1/2)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
Abrar_Lab_1_...	i-097041921de0da7df	Terminated	t2.large	2/2 checks passed	View alarms +	us-east-1e	-	100.49

- Details:** i-097041921de0da7df (Abrar_Lab_1_Server)

 - Instance summary:**
 - Instance ID: i-097041921de0da7df
 - IPv6 address: -
 - Hostname type: -
 - Answer private resource DNS name: -
 - Public IPv4 address: -
 - Instance state: Terminated
 - Instance type: t2.large
 - Private IPv4 addresses: -
 - Public DNS: -
 - Elastic IP addresses: 100.49.166.191 (Abrar_elip_lab_1) [Public IP]